44helper Manual
Every tab in detail

44helper is the node's setup and repair helper: 44Net tunnels, the firewall, the radio software installers, system updates, SD-card care and configuration restore points. This manual goes through it tab by tab and card by card: what each button runs, what a good and a bad result look like, and how to undo it.

STATUS 44NET INSTALLERS UPDATE CONFIG

COVERED BY THIS EDITION

TOOL	FILE	VERSION
44helper	44helper_v0_1_7_20261029.py	0.1.7
SysMon FULL	sysmon_v6_13_85_20261014.py	6.13.85 (Half tunnel checks)

44helper is a common tool: the same file runs on the Pi Zero 2 W and on the full build. The other tools are covered in the user manual, `docs/ASL-DVS_User_Manual.pdf`.

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Markers used in the text

NOTE Background, or something worth knowing.

TIP A quicker or safer way to do something.

CAREFUL This changes your node, restarts something, or can cut you off.

TESTING Built and tested with mocks and in a browser, but not yet proven on many nodes. Expect rough edges and report what you find. Also shown as the badge **TESTING**.

Colours on the page

The Config tab has a Dark / Light switch at the bottom right (from 0.0.224). The choice is kept per browser and per tool; dark is the default.

Plain text is **white**. A step that worked, and a check that passed, is **green**; a failure is **red**; a warning or a check that is still waiting is **amber**; a check that doesn't apply is grey. An output box shows white while a step runs and turns green or red when it reports its result.

Commands

Commands are typed on the Pi, over SSH or at a keyboard and screen. Lines starting with `#` are notes. `<pi-address>` is your Pi's IP address or host name. Example identities use `N0CALL`, `<NODE>` and the documentation addresses `192.168.1.x` and `44.x.x.x`.

PART I

Basics

What 44helper is for, how to install and open it, and how its pages work: step cards, Run and Copy, the checks it makes before it changes anything, and the built-in terminal.

1. What 44helper is, installing it, logging in
2. The page, step cards and the safety checks

What 44helper is

SysMon tells you what is wrong with a node; 44helper is usually where you fix it. It runs as root, on port 9997, and every change it makes is shown as the exact command before you run it.

1.1 What it does

AREA	TABS	CHAPTER
Status	Overview, Boot, Services, Ports, Actions Log; the firewall cards sit on the tunnel tabs	3–5
Remote access	Router Install, Full Tunnel, Split Tunnel, Nodes, Cloudflare	6–9
Installers	ASL3, DVSwitch, SVXLink, M17, Ampersand	10–11
Maintenance	Update, System Optimization	12
Configuration	Config	13

1.2 Installing it

44helper is optional. Install it like the other tools: from instmon's page (Components, or GitHub Updates), with the installer, or by itself:

```
sudo python3 44helper_v0_0_195_20261013.py --install
# and to remove it
sudo python3 /usr/local/bin/44helper.py --uninstall
```

`--install` copies the file into `/usr/local/lib/asl_dvs/44helper/`, links it as `/usr/local/bin/44helper.py`, writes `44helper.service`, which starts through the shared launcher `/usr/local/bin/asl_dvs_launch.py`, and starts it. Its settings live in `/etc/44helper/44helper.conf`; an uninstall leaves them in place.

1.3 Logging in

- Open `http://<pi-address>:9997` and log in with the Pi's **root password**. 44helper has its own login, separate from the Dashboard and SysMon.
- The login **doesn't time out**, because install jobs can run for a long time. It ends when you press **Log out**, when anyone presses **Log out all devices**, or when the Pi reboots.
- Wrong passwords share one 5-try lockout across the login, the terminal and every other password prompt.
- If 44helper restarts while you are logged in, the page reconnects to any running job by itself.

> **CAREFUL** 44helper can install, remove and reconfigure anything on the Pi. Keep port 9997 on your own network. Never open it in a firewall zone or publish it through Cloudflare. The Overview's self-check turns red if it ever becomes reachable through the 44Net zone.

1.4 The header

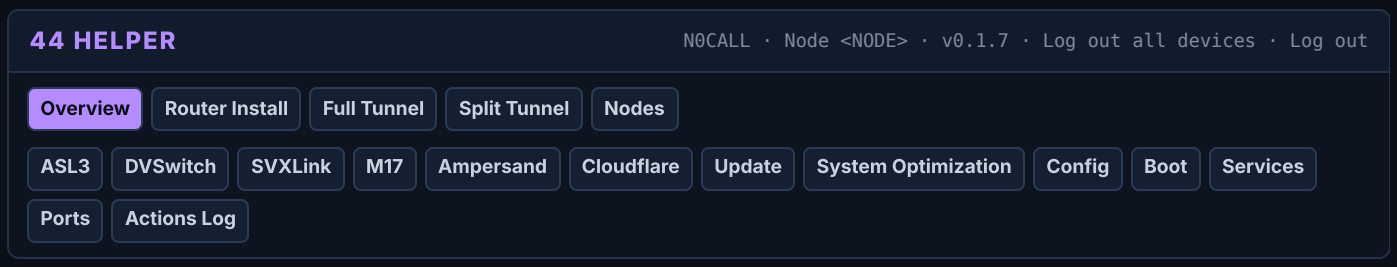


Figure 1.1 · Illustration. The header shows the callsign, node and version. Row one is 44Net remote access; row two is the installers, maintenance, Config and the status tabs.

While quiet mode is on (Update tab, Chapter 12), an amber banner across the top of every tab says so.

The page, step cards and the safety checks

Most tabs are a numbered list of step cards. This chapter explains how a card works, what 44helper checks before it runs a step, and what to do when it refuses.

2.1 A step card

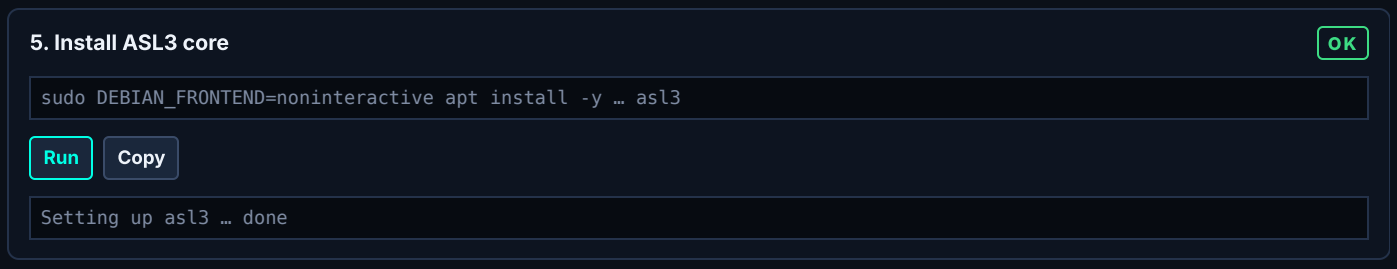


Figure 2.1 · Illustration. Title and status pill, the command, Run and Copy, and the output.

PART	MEANING
Status pill	**Not run**, **Running…** with the elapsed time, **OK** or **Failed**. **Manual** means you run it yourself over SSH (Copy only). **Config** means it saves a Restore point.
Command box	The exact command. You can edit it before pressing Run, for example to change a version or a date. Manual steps are read-only.
Fields	Some steps ask for values (a callsign, a DMR ID, a password, a port). The command is built from them; the field values never appear in the command box.
Run	Runs the command on the Pi as root and streams the output below it.
Copy	Copies the command, and the output if there is any, to run over SSH or send to someone.
▼ note	A folded note at the top of each tab and on some cards. Open it before the first run: it says what changes and what can go wrong.

- Run steps **in order**: each one assumes the ones before it worked.
- Every install tab has a **Script** list (Bookworm, Trixie and others) and a **Custom** choice for typing any one command. The ASL3 list starts on the Debian release the node runs (from 0.1.0); DVSwitch always starts on Bookworm, the only release its packages are published for.
- **Purge previous install** shows the removal steps. After a double confirmation they run one after another and stop at the first failure. Steps marked *Optional* are left out of that pass and only run if you press their own Run button.
- Results last only while the page is open. A reload starts the cards fresh; a job that is still running carries on, and its card picks it up again.

> **CAREFUL** Whatever is in the command box runs as root, edits included.

2.2 Checks before a step runs

CHECK	WHAT HAPPENS
Masked services	A card at the top of each install tab and the Update tab checks whether a service the tab needs is *masked* (blocked from starting). 44helper clears such masks and leaves the service **off**. You can **Turn on** each one, or **Put the mask back (on purpose)**, which 44helper then leaves alone. While a mask can't be cleared, the card is red and the install buttons are locked.
Broken packages	Before any step that installs or removes packages, 44helper runs `dpkg --audit`. If anything is unfinished or broken, the step is refused with **NOT RUN** and a pointer to the Update tab's Package recovery steps.
Package locks	Quiet mode and package steps wait for apt and dpkg to finish; they don't run on top of another package job.

- **What to do next:** when an install step fails, a box opens under it with numbered repair steps. Each has a button that jumps to the right Update-tab card, and a link at the top of the Update tab brings you back.
- **Repair, then purge:** a purge refused because packages are broken offers to repair them in quiet mode and then run the purge again by itself.

2.3 How long a step may run

- **Package and install-script commands** (apt, dpkg, `| bash`, `install*.sh`, builds) have **no time limit**. They run as their own systemd job, so they carry on if 44helper restarts, and the page re-attaches to them.
- Anything else stops after **15 minutes** unless the step sets its own limit.
- Pi tunnel step 6 waits up to **90 seconds** for the tunnel to start (Chapter 7).

2.4 The built-in terminal

A few steps need you to answer questions on screen: `asl-menu`, the USRP2M17 installer and Ampersand's first run. Their cards can open a real terminal in the page, but only:

- after it has been **switched on**, from the page with the root password, or by creating `/etc/44helper/terminal.enable` over SSH (owned by root, not writable by others);
- from your local network, never through a tunnel or Cloudflare;
- with the root password entered again each time you open it;
- for the one command printed on that card, nothing else.

One terminal at a time. It closes after 10 minutes with no typing, or when you log out.

PART II

Status tabs and firewall

Five tabs that read the node and change little: Overview, Boot, Services, Ports and Actions Log. Then the firewall cards, which from 0.0.196 sit at the end of the Full and Split Tunnel tabs instead of a Firewall tab of their own, and the half tunnel card on the Full Tunnel and Split Tunnel tabs (from 0.0.214).

3. Overview, Boot, Services and Ports
4. Firewall and port control, and Actions Log
5. The half tunnel card

Overview, Boot, Services and Ports

3.1 Overview tab

The banner at the top sums up the worst finding. The cards below it:

CARD	SHOWS
Installed software	ASL3, DVSwitch, USRP2M17, SVXLink and Ampersand. Each is **installed** (version and running services), **broken** (half-installed packages, or a package without its program files), **leftovers** (removed, but settings, folders or unit files remain) or **absent**. Click a name to open its tab.
Tunnel / wg0	Whether `wg0` is up, its address and peer, and the last handshake.
AllStar registration	From 0.0.212. Each node from `rpt show registrations`, green when registered and red when not, with the server it registers with and the address and port the AllStarLink server sees it at (the Perceived column): **Registered as 44.x.x.x:4569 · through the 44Net tunnel**, **your public internet address (not the tunnel)**, or **a private address (check NAT)**. On a split tunnel your public address is normal; on a full or half tunnel it should be the 44 address.
Configured tunnels	The 44Net tunnels 44helper has set up: on the Pi (full or split) or on a router.
44NetConnect firewall zone	The services and ports open to the 44Net side.
Self-check findings	**Red** if 44helper, SysMon, the Dashboard, Cockpit or Asterisk AMI is open in the 44NetConnect zone. **Amber** if `wg0.conf`, which holds your private key, is not mode 600. From 0.0.197: **red** when the tunnel server routes back into wg0 (a routing loop, section 7.3), **amber** when it is inside AllowedIPs and the endpoint pin is missing. From 0.0.198 four more checks, each from a problem seen on a real node: • **Tunnel handshake**: green under 3 minutes old; **red** when older with a `PersistentKeepalive` set, or never (the 44Net Connect dashboard's "idle"); **amber** when older with no keepalive, since a quiet tunnel may only be idle. • **Services that didn't start**: **red** when netcheck failed, listing the DVSwitch services that never started; **red** for any other failed service; **amber** for an enabled DVSwitch service that isn't running. • **DNS line that can't work**: **red** when wg0.conf's DNS setting needs systemd-resolved and resolved doesn't answer within 5 seconds, or there is no `resolvconf`. The next tunnel start would fail. • **Address not a /32**: **amber**, naming the network the wider prefix routes into wg0.

Re-check reads everything again at once; otherwise the installed-software list is cached for 20 seconds, and the four checks above run at most every 30 seconds. From 0.0.199 a finding that has a fix shows **Fix on the tunnel tab →**: it opens the Full or Split tab, whichever is installed, and highlights that row of the Health and fixes card (section 7.6).

3.2 Boot tab TESTING

From 0.1.0 the Overview also shows the system it runs on, under the Re-check button: the board (Raspberry Pi or not), the OS and Debian release, the processor and the WiFi manager.

From 0.0.209, between Config and Services: what happened when the node last started, in one place, to read, copy, or hand to someone helping you (or an agent). 44helper serves both builds, so the tab is on the full build too. It is read-only, except **Save a copy now**.

This boot

The summary: boot time and uptime, this boot's line from `/var/lib/44helper/boot.log` (7.7), where the journal is kept (disk, or memory only) and how many boots it holds, the clock (NTP on, synchronised), the Pi's power and temperature limits (`vcgencmd get_throttled`), the boot order, the **Tunnel** line, wifimon's newest shutdown report and tunnel watchdog state, and every tool's version. The Tunnel line says **policy routing (table 4445)** on a split tunnel set up the 0.0.212 way (7.3), whether the tunnel server is on the normal route or routed into wg0, and warns while the older routing is still in use.

Below it, a **findings** list names what went wrong:

FINDING	FROM
netready not ready in time; the late follow-up ran	0.0.209
Failed units; units systemd restarted; enabled services that never started (a unit they require failed)	0.0.209, 0.0.211
Clock not synchronised; `systemd-time-wait-sync` holding the clock target	0.0.209, 0.0.211
Under-voltage; an unclean stop before this boot	0.0.209
The tunnel watchdog (wifimon) acted or gave up	0.0.209
The boot order incomplete, and the Health card's warnings	0.0.209
A routing loop (tunnel server routed into wg0); the older tunnel-server route line	0.0.210
ASL3 updates (`asl3-update-astdb`, `asl3-update-nodelist`) that took minutes	0.0.211
Asterisk bound to the tunnel address, and up before the tunnel	0.0.211
An "ordering cycle" logged by systemd at this boot	0.0.212

Timeline

Read from `systemctl show`, so it works with a journal kept in memory. For each unit, the seconds after boot that it started and became active, with its result, restarts and enabled state, and a **Last started** column (the latest start, so a unit restarted later shows that time). Rows: time sync and `time-wait-sync`, wait-online, the tunnel, netready and its late follow-up, netcheck, the DVSwitch services, Asterisk, the ASL3 update units, the Node Controller tools and `multi-user.target`. **Red**: failed. **Amber**: restarted by systemd, or the late follow-up ran.

Logs

Each log is read only when you press its **View** or **Copy** button:

- the boot log; this boot's journal for the tunnel, netready, netcheck and DVSwitch (`-o short-monotonic`); the previous boot's, when the journal keeps it;
- `systemd-analyze` (time, the critical chain of the tunnel and netready, blame); the clock; kernel lines for WiFi (including USB WiFi drivers), link, power and storage, with repeated lines shown once with a count; power limits;
- the tunnel now: `wg show`, the routing rules and routes, and the route to the tunnel server with WireGuard's fwmark;
- wifimon: watchdog status, lost-network reboots, its tunnel and shutdown lines, the newest shutdown report;
- the Dashboard's boot tune and disconnect lines; instmon's held backups.

Saved copies of earlier boots

The journal is often kept in memory only (System Optimization, 12.3), so the previous boot is gone after a reboot. 44helper saves the boot report to `/var/lib/44helper/boots/` **6 minutes** after each boot and updates it at **25 minutes** (after the late follow-up), one file per boot, the **newest 5** kept, at most 64 KB each, mode 0600. Each one has View, Copy and Download. **Save a copy now** updates this boot's file.

Copy boot report and Download .txt

One plain-text report: a header with the host, the boot and the versions, then the findings, the timeline and every log, times in seconds after boot, at most 60 KB. **Secrets are removed**: PrivateKey and PresharedKey lines, `password=` and `token:` style values, `--token` arguments and long token strings. It is meant for pasting into a help request.

For agents and scripts

```
# logged in to the page (the login cookie)
GET /api/boot/report?format=json   # keys: summary, findings, timeline, sections, snapshots
GET /api/boot/report?format=text   # the text report
# over SSH
sudo python3 /usr/local/bin/44helper.py --boot-report [--json]
```

Also `/api/boot/status`, `/api/boot/section?id=…` and `/api/boot/snapshot?name=…`. The tab works without the other tools' newer versions: a missing boot log, tunnel watchdog status, Dashboard or instmon shows "not there" instead of failing.

3.3 Services tab

Grouped lines, one per service or listening port. Masked services are marked.

GROUP	SERVICES
System	firewalld, systemd-resolved
44Net Tunnel	`wg-quick@wg0`, or the ARDC fallback unit `44net-tunnel.service`
AllStarLink / DVSwitch	Asterisk, MMDVM_Bridge, Analog_Bridge, md380-emu, and the gateway command ports
M17	usrp2m17
SVXLink	svxlink
Ampersand	amp-server, amp-hub
Cloudflare Tunnel	cloudflared
Dashboards (should stay private)	the Dashboard (`nodecontrol`, or `asl_dvs_dashboard` on older installs) and `dvs_dashboard`

3.4 Ports tab

A reference table of every port the node's software uses, checked against what is reachable through your remote-access setup. The line above it says where it looked: the Pi's 44NetConnect zone, the router's forward table, or "none (no tunnel configured yet)". **Running** means something listens on the port (for Cloudflare's outbound row, that `cloudflared` is up).

PORT	USED BY	ROLE
4569/udp	AllStarLink IAX2	peer-inbound
5198-5199/udp	EchoLink (Asterisk or SvxLink)	peer-inbound
1667/udp	VOTER/RTCM	peer-inbound
4569/udp	Ampersand hub IAX (form default; set in the hub's step 4)	peer-inbound
5038/tcp	Asterisk AMI	mgmt-LAN
8989, 9999/tcp	Dashboard, SysMon	mgmt-LAN
9090/tcp	Cockpit	mgmt-LAN
8080/tcp	Ampersand web page	mgmt-LAN
31100, 31103/udp	Analog_Bridge ↔ MMDVM_Bridge	loopback-only
32008, 34008/udp	USRP2M17 ↔ Asterisk	loopback-only
32012, 34012/udp	Zello bridge	loopback-only
6073, 6074, 6075/udp	YSF, P25, NXDN gateway commands	loopback-only
54321/udp	ircDDBGateway remote control	loopback-only
7844/udp	Cloudflare Tunnel edge	outbound-only

VERDICT	MEANS
OK	As it should be.
over-exposed	A mgmt-LAN, loopback-only or outbound-only port is open to the outside. This is the security finding.
needed-but-not-open	A peer-inbound port is running but not open, so inbound links will fail. Only shown once a tunnel is set up.
not-listening	Nothing runs on it: normal for software you don't use.

NOTE "Exposed" means in the 44NetConnect zone (Pi tunnel), in the router's forwards (router, single address), or on the whole subnet (router, routed subnet with masquerade off). A Cloudflare Tunnel is never visible here: nothing opens inbound for it (Chapter 9).

Firewall and port control, and Actions Log

44helper uses **firewalld**. The **Firewall and port control** section shows what each zone lets in and changes it, one port or service at a time, with a safe "try it first" path.

4.1 Where it is

Before 0.0.196 these cards had a Firewall tab. Now they are a section at the end of the **Full Tunnel** and **Split Tunnel** tabs, under the Purge buttons. The section exists once and moves to whichever of the two tabs you open, so a port you typed or a zone you picked stays as it was, and opening the tab reads the firewall again. It sits outside the tab's step list, so it stays usable while the Split steps are locked. The other two cards moved too:

CARD	NOW ON
Half tunnel (ASL + EchoLink only)	Full Tunnel and Split Tunnel tabs, above Full tunnel with exceptions (Chapter 5)
Router: live port-forward table	Router Install tab, under the steps (Chapter 9)

4.2 The cards

CARD	WHAT IT DOES
Guardrail	Warns when `wg0` is attached to the ASL3 image's own `allstarlink` zone instead of 44helper's `44NetConnect` zone.
Pi: 44NetConnect zone	What the 44Net side can reach. Add or remove the services `iax2`, `echolink` and `rtcm`, or a port such as `14569/udp`.
Quick port control (any zone)	Section 4.3.
Panic mode	If firewalld's panic mode (all traffic dropped) is on, a button turns it off. 44helper never turns it on.
Re-check	Reads the zones and the guardrail again; "Last checked" shows when.

4.3 Quick port control

1. Pick a zone. The 44NetConnect zone is pinned at the top; the active zones show their interfaces, and the default zone is marked. **Show zone config** lists everything in it (read-only).
2. Type a port such as `4569/udp`, or press a preset: IAX2 4569/udp, EchoLink/SvxLink 5198-5199/udp, VOTER/RTCM 1667/udp, or your DVSwitch Mobile port.
3. Press **Add (runtime test)**. It opens the port until the next firewalld reload, without touching the saved config.
4. When it works, press **Make permanent**, which saves that one port and reloads. **Remove** closes it everywhere (runtime and saved).

BUTTON	RUNS
Add (runtime test)	`firewall-cmd --zone=<z> --add-port=<p>`
Make permanent	`firewall-cmd --permanent --zone=<z> --add-port=<p> ; --reload`

Remove	the same with `--remove-port`, runtime and permanent, then `--reload`
Commit all runtime changes	`firewall-cmd --runtime-to-permanent`
Reload firewalld	`firewall-cmd --reload`
Advanced	add or remove a service in a zone (checked against `--get-services`), move an interface to a zone (`--change-interface`) or take it out

- **Guardrail as you type:** a loopback-only port (Chapter 3) is **blocked** in every zone, by the server and not just the page. A dashboard, AMI or Cockpit port is allowed but flagged red.
- Every zone name is checked against firewalld's own list, every port against `port/proto`, and interface and service names against a strict pattern.
- **During a half-tunnel test**, Commit all runtime changes is refused (it would save the test's temporary ports), and Reload warns that it closed them.

> **TIP** Always use Add (runtime test) first. If you lose access, a reload or a reboot puts things back. Make it permanent once you know it works.

4.4 Actions Log tab

- **Session log:** every command 44helper ran since it last started, with OK or FAIL. **Refresh** reads it again. A step that puts a password in its command (such as Allmon3's) shows it here too.
- **Known-good router notes:** notes saved after a router setup worked (model, firmware and access method, no secrets). Click one to read it.

The half tunnel card TESTING

A full tunnel makes your 44 address reachable by anyone, but sends all the Pi's traffic through 44Net. A split tunnel keeps the Pi's traffic at home, but only 44Net hosts can reach you. The half tunnel takes the useful half of each: anyone can reach ASL and EchoLink at the 44 address, and everything else keeps the normal route.

5.1 What changes

TRAFFIC	GOES
Asterisk (AllStar links, registration, EchoLink)	through the tunnel, from the 44 address
Anything sent from the 44 address	through the tunnel
The IAX port and EchoLink 5198-5199/udp, 5200/tcp	through the tunnel
Traffic to 44Net, the LAN, and the tunnel server itself	its normal route
DVSwitch, usrp2m17, DNS, updates, everything else	the normal internet link

How: wg0 runs with `Table = off` and `AllowedIPs = 0.0.0.0/0`, so WireGuard accepts packets from any address but adds no routes. A route table (4444) with a default route via wg0 and six routing rules (priority 1100–1105) pick out Asterisk's traffic. The first rule, `lookup main suppress_prefixlength 0`, keeps the LAN, 44Net and the tunnel server on their normal routes.

> **CAREFUL** With the half tunnel on, other nodes and EchoLink reach you at the 44 address only. A router port forward to the IAX or EchoLink ports stops working, because Asterisk's replies go out the tunnel from the 44 address. Everything open in wg0's firewall zone is reachable from the whole internet.

5.2 Before you start

- The Full Tunnel or Split Tunnel tab done, with the tunnel working. A split config is fine: the test uses a copy set to `0.0.0.0/0`.
- Asterisk running as its own `asterisk` user (ASL3 does). As root, only the IAX and EchoLink ports use the tunnel, and registration still shows your home address.
- Do it from home, on the LAN, not over the tunnel.

5.3 The buttons

#	BUTTON	WHAT IT DOES
0	Check first	Read-only. Finds the 44 address, the tunnel server, the IAX port (from `iax.conf`), whether EchoLink is loaded, the normal route, the LAN, wg0's zone and Asterisk's user. Backs up wg0.conf to the instmon library as `wg0.conf.pre-halftunnel`. Blocks on anything that would break (5.4).
1	Arm auto-undo	A 15-minute systemd timer that runs `/run/44helper-half/undo.sh`, even if the page or SSH drops. Nothing below runs until it is armed. Press it again to restart the 15 minutes.

2	Restart tunnel in half mode	Stops `wg-quick@wg0` and brings wg0 up from a copy in `/run` with `Table = off` and `AllowedIPs = 0.0.0.0/0`. Checks the normal default route is untouched. Before it starts, the DNS check from Chapter 7 runs.
3	Add tunnel route table	Table 4444 with a default route via wg0, plus the 44Net routes (44.0.0.0/9, 44.128.0.0/10). A tunnel server inside 44Net is pinned to the normal route.
4	Send ASL and EchoLink into the tunnel	The routing rules: main without its default (1100), from the 44 address (1101), Asterisk's user (1102), the IAX source port (1103), EchoLink 5198-5199/udp (1104), EchoLink directory 5200/tcp (1105).
5	Open tunnel ports	The IAX port and EchoLink, runtime only, in wg0's zone. Ports already open are left alone.
6	Loosen the return-path check on wg0	`rp_filter = 2` on wg0. The kernel uses the higher of `all` and `wg0`, so this is enough.
	Run 2–6 in order	Runs the steps it hasn't done yet and stops at the first failure, putting everything back. A failed step puts back its own part first.
7	Run checks	14 read-only checks (5.5).
8	Undo now	Runs the undo script now: newest change first, carrying on past a stuck piece, logging to `/run/44helper-half/undo.log` (shown on the card).
9	Keep for now	Cancels the timer. The test stays until Undo now or a reboot. The step buttons then need the timer armed again.
10	Make permanent	5.6.
11	Remove permanent	5.7. Double confirm.

> **NOTE** Nothing in the test is written to disk except under `/run`, so a reboot also clears it.

5.4 What Check first blocks on

- no wg0.conf, no 44 address or no `[Peer]`; a tunnel server it can't find or look up;
- a `Table =` line already in wg0.conf, or `SaveConfig = true` (wg-quick would rewrite the file and drop the markers);
- an unusual AllowedIPs (neither full nor split);
- firewalld not running, wg-quick missing, the ARDC fallback unit running, an `ip` command too old for `uidrange`, `sport` and `suppress_prefixlength`;
- no normal default route, or one pointing into the tunnel; wg0 in no firewall zone; Full Tunnel outbound block rules loaded;
- wg0 up but not from `wg-quick@wg0`; the full tunnel with exceptions in use (Chapter 8); 44helper not running as root.

5.5 The 14 checks

Each line is **green** (pass), **red** (fail), grey (doesn't apply) or **amber** (waiting). Registration and EchoLink stay amber for up to 6 minutes while they update. Check first records how checks 11–14 looked before the test, so a failure that was already there is labelled that way.

#	CHECK	PASSES WHEN
1	Tunnel is up	The 44Net server was heard from in the last 3 minutes.
2	Running in half mode	wg0 is up with `Table = off` and none of wg-quick's full-tunnel rules.
3	No conflicting rules	No outbound REJECT/DROP direct rules are loaded.
4	Internet still uses the normal link	`1.1.1.1` does not route via wg0.
5	DNS works	`allstarlink.org` resolves.
6	LAN tools still respond	The Dashboard, SysMon, instmon and 44helper ports that were listening still answer, their replies to the LAN gateway leave by the LAN link, and replies to your browser don't go out wg0.
7	Tunnel route table exists	Table 4444 has its default route, and the 44Net routes are on wg0.
8	Routing rules are loaded	Every rule from step 4 is in place.
9	ASL traffic really goes into the tunnel	A test lookup for the IAX port, Asterisk's user and EchoLink comes back via wg0, from the 44 address.
10	Ports open on the tunnel only	The ports are open in wg0's zone and the return-path check is loose.
11	DVSwitch connected	The bridges run, and the newest MMDVM_Bridge log shows a DMR login.
12	usrp2m17 connected	The service runs.
13	ASL registration shows your 44 address	`rpt show registrations` lists the 44 address.
14	EchoLink logged in	`echolink dbdump` has directory entries.

A red line on check 4, 5 or 6 shows a banner recommending **Undo now**: those are the ones that take the node off the air.

5.6 Make permanent

Unlocks when steps 2–6 are done and the last checks, run after them and under 15 minutes old, are all green or grey. Type the user whose home folder gets the removal script (default: the UID 1000 user; root or a login user, UID 1000 and up, only). Then it:

1. Saves each piece in its normal home: wg0.conf gets `Table = off`, `AllowedIPs = 0.0.0.0/0` and the routes and rules as PostUp/PostDown lines between `# 44helper-half-begin` and `# 44helper-half-end` (they work out the gateway at start-up, so a travel node on another network still works); firewalld gets the ports in wg0's zone; `/etc/sysctl.d/90-44helper-halftunnel.conf` gets the return-path setting.
2. Writes `remove_half_tunnel.sh` and a 0600 copy of the original wg0.conf into that home folder and `/root`, and a login reminder in `/etc/profile.d/`.
3. Clears the test, starts the tunnel from the saved wg0.conf, reloads the firewall and runs the checks again.
4. If a check fails, it takes everything back out and restarts the test with a fresh 15-minute timer. If a check is only waiting (registration), it stays saved.

5.7 Remove permanent

Runs the same steps as `~/remove_half_tunnel.sh`. The script is plain bash, needs no 44helper and asks for sudo by itself. It stops the tunnel, removes the marked lines and puts AllowedIPs back (so any other edits to wg0.conf are kept), deletes the rules and routes, closes the ports it opened, removes the sysctl file and starts the tunnel again. If the marked lines can't be taken out, it restores from a backup, root-owned copies first.

> **NOTE** While the half tunnel is being tested or is permanent, the Split tab's AllowedIPs repair is switched off: narrowing AllowedIPs would cut off every node outside 44Net. SysMon 6.13.84 (full build) has the outside-view checks too: Console → Troubleshooting → Half tunnel.

PART III

44Net and remote access

How to make the node reachable, or reach it yourself while away: a 44Net tunnel on the Pi (split, full, half, or full with exceptions), a 44Net tunnel on your router, or a Cloudflare Tunnel for the web pages.

6. Choosing a route
7. Full Tunnel and Split Tunnel tabs
8. Full tunnel with exceptions
9. Router Install, Nodes and Cloudflare

Choosing a route

44Net (AMPRNet) gives amateur stations real public 44.x.x.x addresses. 44Net Connect is ARDC's free WireGuard service that hands you one. How much of the Pi uses it is up to you.

6.1 The four Pi tunnel modes

	SPLIT	HALF	FULL WITH EXCEPTIONS	FULL
Where it is	Split Tunnel tab	Full Tunnel tab card	Full Tunnel tab card	Full Tunnel tab
AllowedIPs	`44.0.0.0/9, 44.128.0.0/10`	`0.0.0.0/0`	`0.0.0.0/0`	`0.0.0.0/0`
Pi's internet traffic	normal link	normal link	tunnel	tunnel
What uses the tunnel	traffic to 44Net	44Net, plus ASL and EchoLink	everything except LAN replies and a bypass list	everything
Who can reach the 44 address	44Net hosts only	anyone	anyone	anyone
What registration shows	home address	44 address	44 address	44 address
Router forward for ASL	needed	not needed	not needed	not needed
Works behind CGNAT	no (for ASL)	yes	yes	yes
Router-forwarded dashboard/SSH	works	works	works (step 2)	hangs

Why split only lets 44Net in: a packet from an ordinary internet address does arrive down the tunnel, but WireGuard checks its source against `AllowedIPs` and drops it. That is why a split tunnel keeps your home address and router forward for ASL.

Half vs full with exceptions: both let anyone reach the 44 address. They differ in what happens to traffic that isn't on a list. With the half tunnel it stays on the normal link, so a new service or an update can't break by surprise. With full with exceptions it goes through the tunnel, so the whole Pi shows up as the 44 address, and each service that misbehaves needs an exception.

6.2 Which to pick

- **Split:** your ISP gives you a public address with a router forward for ASL, and you want 44Net peers too. The lowest-risk mode.
- **Half:** ASL should be reachable at the 44 address with no router forward, or you're behind CGNAT, and the rest of the Pi should be left alone. The safest way to do that on a DV node.
- **Full with exceptions:** the whole Pi behind the 44 address, but forwarded dashboard and SSH connections should still answer.
- **Full** on its own: a Pi that is only reached through 44Net.

6.3 Router and Cloudflare

ROUTE	TAB	BEST FOR
Router tunnel	Router Install	Several devices, or a whole routed 44Net subnet, behind one tunnel on an OpenWrt / GL.iNet router.
Cloudflare Tunnel	Cloudflare	Opening the web pages yourself from anywhere, with no port opened at home and a Cloudflare login in front. It carries web pages only: it can't carry IAX or EchoLink voice.

TIP A split tunnel and Cloudflare make a good pair. The tunnel handles radio traffic with 44Net; Cloudflare lets you check the pages from your phone without opening them to 44Net or the internet. Under a full tunnel, cloudflared's own connection goes through ARDC too.

NOTE With a full, half or exceptions tunnel, the Pi's traffic crosses ARDC's network. Check that what you send fits ARDC's 44Net acceptable use policy.

Full Tunnel and Split Tunnel tabs

From 0.1.2: step 1 (firewalld) stops when ufw is the active firewall, because two firewalls fight over the tunnel's rules. On Debian, installing systemd-resolved can leave names unresolved on a machine without NetworkManager; step 1a now sets the DNS servers the machine used before. Start stops with a message when neither `nft` nor `iptables-restore` is installed, instead of taking the tunnel down. Step 1a installs `wireguard-tools` without its recommended packages, so apt doesn't install a new kernel.

Both tabs set up WireGuard (`wg0`) on the Pi and a firewalld zone called **44NetConnect** that decides what the 44Net side can reach. The steps are the same; the only difference is the `AllowedIPs` line in the config you paste.

44Net Connect gives every tunnel the same config, with `AllowedIPs = 0.0.0.0/0`. The Split tab needs `44.0.0.0/9, 44.128.0.0/10` instead. From 0.0.200 you paste the config as it is: step 4's check offers to rewrite that line. A 44Net Connect tunnel is a single address, not a routed subnet; for a subnet use the Router Install tab.

From 0.0.200 the install also prevents the problems the Health and fixes card (section 7.6) repairs: step 1a checks that systemd-resolved answers, step 4 tidies the config before writing it, and step 6 finishes the job after the start.

7.1 The steps

#	STEP	WHAT IT DOES
1	Firewall prereq	Installs and starts firewalld.
1a	systemd-resolved prereq	Installs WireGuard and systemd-resolved, without which the tunnel can fail to start. From 0.0.200 the step is green only when resolved is running **and** answers (`resolvectl status` within 5 seconds); the card shows Running and Answers separately. Running the step restarts resolved once if it isn't answering. If it still doesn't answer, the step stays red: reboot, and if that doesn't help, set DNS to Off at 4a.
2	Create the 44NetConnect zone	`/etc/firewalld/zones/44NetConnect.xml`.
3	Inbound services and ports	Tick what other stations may reach: **AllStarLink IAX2** and **EchoLink / SvxLink** (on by default), **VOTER/RTCM** (optional). Asterisk AMI can be added but is flagged red. The Full tab also has your DVSwitch Mobile port and a custom port. From 0.0.212 the Split tab offers IAX2 and VOTER/RTCM but not EchoLink: only 44Net hosts can reach a split tunnel. If firewalld has no `iax2` (or `echolink`) service, the step opens the ports instead: the IAX port from `iax.conf` `bindport`, default `4569/udp`. The step counts as done only when IAX is open.
4	Paste WireGuard config	Paste the whole `[Interface]` / `[Peer]` block and press **Check and write wg0.conf (chmod 600)**. A config with no real private key is refused. A panel shows what the check will change (below); nothing is written until you press **Write wg0.conf**. Leave **Also back up a copy to the instmon library** ticked so you can recover the tunnel after re-flashing the card. **Load from instmon library** puts a saved copy back (it shows your private key). From 0.0.212, once wg0.conf exists the step shows **Replace the config** instead (a new key or tunnel server): the same check before writing, the DNS and MTU settings carried over, and the old file saved as `/etc/wireguard/wg0.conf.pre-replace`.
4a	DNS & MTU	7.2.

4b	Split: Split tunnel routing · Full: reflector ports	7.3.
5	Attach wg0 to the zone	Moves wg0 into 44NetConnect.
6	Enable + start the tunnel	7.4.

The check before writing (step 4, from 0.0.200)

IN THE PASTED CONFIG	WHAT HAPPENS
`SaveConfig = true`	Removed, always. wg-quick would rewrite the file every time the tunnel stops and drop 44helper's marked lines.
A `DNS =` line	Removed while systemd-resolved isn't answering, or when there is no `resolvconf`, so DNS is Off and the start can't fail on it. Set DNS on card 4a once step 1a is green. Kept when resolved answers. From 0.0.212 a DNS server outside AllowedIPs is dropped from the line, with a note: lookups sent to it over wg0 would be dropped. A 44Net server such as 44.0.0.1 is kept.
One 44Net address wider than /32	**Change Address to /32**, ticked by default. Untick it only if ARDC gave you a subnet; check the portal.
No `PersistentKeepalive`	**Add PersistentKeepalive = 25**, ticked by default. Without it a quiet tunnel goes idle and stations on 44Net can't reach the node.
AllowedIPs that don't fit the tab	Two choices: **Rewrite it** to this tab's shape (selected), or **Stop** and paste it on the other tab. Without a choice nothing is written. From 0.0.212 the Split tab wants exactly `44.0.0.0/9, 44.128.0.0/10`: `0.0.0.0/0`, `::/0`, a pair of `0.0.0.0/1` halves and `44.0.0.0/8` (which takes in addresses that aren't amateur radio's) all get the same rewrite offer.
A split tunnel	From 0.0.212 the policy-routing block is added (7.3), and an older tunnel-server route line is removed. The panel shows it.
Tunnel server inside AllowedIPs, no policy routing	The endpoint pin is added (7.3). This is now only for configs that can't take the policy-routing block.

- Convert's write step runs the same check, and so does a config loaded from the library (it fills the paste box).
- Every other write of wg0.conf (cards 4a and 4b, the Health fixes, the half tunnel and the exceptions) also removes `SaveConfig = true` and says so.
- The check sends back only its notes and choices, never the config or its key.

7.2 Step 4a: DNS and MTU

DNS OPTION	WRITES INTO WG0.CONF	DEFAULT SERVERS
Off	removes 44helper's DNS lines; the Pi keeps its own resolver	—
All DNS through the tunnel	`PostUp = resolvectl dns %i …; resolvectl domain %i ~.`	1.1.1.1, 9.9.9.9
Only 44Net names	`PostUp = resolvectl dns %i …; resolvectl domain %i ~ampr.org ~44.in-addr.arpa`	44.0.0.1

A `DNS =` line	`DNS = …`, which wg-quick hands to `resolvconf`	1.1.1.1, 9.9.9.9, 44.0.0.1

- The lines go between `# 44helper-dns-begin` / `-end` markers. Off removes only those; a `DNS =` line that came with the pasted config stays.
- From 0.0.201 the resolvectl options undo themselves with `PreDown = resolvectl revert %i`. Earlier versions used PostDown, which wg-quick runs after wg0 is deleted: it failed and stopped the rest of the shutdown. Older files are changed on the next write, step 6 or Restart tunnel, and the Health card offers **Move to PreDown**.
- From 0.0.201 a DNS server outside the tunnel's AllowedIPs is refused, because lookups sent to it over wg0 are dropped. On a split tunnel use "only 44Net names" with a 44Net server such as 44.0.0.1, or Off.
- From 0.0.193 the resolvectl options, and the `DNS =` line when `resolvconf` is systemd-resolved's, are refused while systemd-resolved isn't answering.
- **MTU:** a number from 1280 to 1500 (blank or `off` removes it), between `# 44helper-mtu-begin` / `-end`. WireGuard's own default is 1420.
- **The cards own their settings** (0.0.212). Setting DNS removes any raw `DNS =` line, and setting MTU removes any raw `MTU =` line, so the value on the card is the one in force.
- From 0.0.211 the `PreDown = resolvectl revert %i` line ends in `|| true`. Without it, when systemd-resolved didn't answer, wg-quick stopped at that line and left wg0 up, so every later start failed. Older lines are fixed on the next write, by the Health card, Restart tunnel or Set the boot order.
- Restart the tunnel for either to take effect: `sudo systemctl restart wg-quick@wg0`.

> **TIP** With a full tunnel the Pi's normal DNS lookups already go through the tunnel, so Off is usually fine. A `DNS =` line is the most fragile option: it depends on `resolvconf`, and on these Pis that means systemd-resolved over D-Bus.

7.3 Step 4b: Split tunnel routing

The problem. Some 44Net Connect servers are themselves 44.x addresses. With a split tunnel, wg-quick sends all of 44.0.0.0/9 into wg0, the server included, so WireGuard's own packets loop back into the tunnel: no handshake, almost nothing received, and the 44Net Connect dashboard says **idle**. `ip -4 route get <endpoint>` shows `dev wg0`. Up to 0.0.211 44helper fixed this with a route copied when the tunnel starts (the endpoint pin, below). That route belongs to one adapter: when WiFi moved to another adapter, or Ethernet came and went, the kernel dropped it and the tunnel looped again.

Policy routing (0.0.212). A split tunnel (every AllowedIPs inside 44/8, IPv4 only, no `Table` or `FwMark` of its own) gets a routing block in wg0.conf, between `# 44helper-routing-begin` / `-end`:

```
Table = 4445
FwMark = 0x4445
PostUp = ip -4 rule del priority 10044 2>/dev/null || true; ip -4 rule del not fwmark 0x4445 table
4445 priority 10045 2>/dev/null || true; ip -4 rule add not fwmark 0x4445 table 4445 priority 10045
PostDown = ip -4 rule del not fwmark 0x4445 table 4445 priority 10045 || true
```

The 44Net routes go into table 4445, and one rule (priority 10045) sends everything *without* the mark to that table. WireGuard marks its own packets, so they skip table 4445 and take the normal route, on whichever adapter is up. The tunnel server can't loop back into wg0, whatever the network does. The tunnel-server route line is then removed, and no NetworkManager hook is needed. A full tunnel is unchanged. Every 44helper write of wg0.conf applies the block, and the paste check (7.1) shows it.

CARD 4B SHOWS	MEANS	BUTTON
Set	Policy routing is in wg0.conf and in the running tunnel.	—
Restart needed	The block is in wg0.conf, but the running tunnel still uses the old routing.	**Restart the tunnel with the new routing** (from the LAN)
Older routing	The split tunnel still uses the tunnel-server route line; an amber banner says so, and whether it loops right now.	**Switch to policy routing**: writes the block, restarts the tunnel (a LAN session stays up), waits for a handshake and checks the rule is there.

Set the boot order (7.7) also converts an older split tunnel ("split tunnel routing (older form)" in what is missing) and then says to restart the tunnel. The Boot tab's Tunnel line (3.2) says **policy routing (table 4445)**, and warns while the older form is in use.

> **NOTE** 0.0.212 used table 4444, mark 0x4444 and priority 10044, which is also the half tunnel's table (Chapter 5). From 0.0.213 the split tunnel uses its own table 4445 (mark 0x4445, priority 10045); a file written by 0.0.212 counts as the older form and is rewritten, and the new PostUp line removes the old rule.

The older form: endpoint pin (0.0.197–0.0.211)

Kept only for split tunnels not converted yet, and for configs that can't take the block (their own `Table`, or IPv6). Two lines between `# 44helper-endpoint-pin-begin` / `-end`:

```
PreUp = (wait up to 30 s for the normal default route); ip -4 route replace <server>/32 <that route>
PostDown = ip -4 route del <server>/32 || true
```

- From 0.0.206 the PreUp line waits up to 30 seconds for the normal default route; if there is none, the start fails and the boot order's retry (7.7) tries again 20 seconds later. A host-name endpoint is looked up then and saved in `/run/44helper-endpoint-wg0`.
- Card 4b's red banner and **Fix the tunnel-server route** add it and the route at once; an amber banner and **Update the tunnel-server route** rewrite a pin from before 0.0.206.
- 0.0.210 added a NetworkManager hook, `/etc/NetworkManager/dispatcher.d/90-44helper-endpoint`, that moves the pinned route to the current default route when an interface comes up or goes down, DHCP renews or connectivity changes (`journalctl -t 44helper-endpoint`). From 0.0.212 the boot order needs it only while a pin is left.
- From 0.0.211 a write while DNS is down no longer removes the pin of a host-name endpoint.

Split: Repair AllowedIPs is on the same card when the config still has the full-tunnel AllowedIPs. Under a split tunnel a full config sends every DNS lookup into the tunnel, the classic cause of "AllStar works but DVSwitch and M17 don't". It rewrites AllowedIPs to `44.0.0.0/9, 44.128.0.0/10`; restart the tunnel after. It is switched off while the half tunnel or the full tunnel with exceptions is in use.

Full: reflector ports adds firewalld direct rules on the OUTPUT chain, scoped to `-o wg0`: outbound allows for the digital-voice reflectors the node dials out to, never an inbound opening. They only matter if outbound traffic is restricted.

MODE	PORT	STATUS
M17 · YSF · P25 · NXDN	17000 · 42000 · 41000 · 41400	confirmed
D-STAR DPlus · DExtra · DCS	20001 · 30001 · 30051	confirmed

MMDVM / XLX	`62030`	confirmed
BrandMeister DMR	`62031`	opt-in; confirm against your server
DMR+ / HBlink	`55555`	opt-in; can collide with other DMR back-ends
XLX / URF interlink	`10002 · 10017`	opt-in; confirm against your reflector

7.4 Step 6: Enable + start the tunnel

Runs `systemctl enable wg-quick@wg0` and `systemctl start wg-quick@wg0` (or the ARDC fallback unit `44net-tunnel.service` where the wg-quick template is missing). From 0.0.212, if the tunnel is already running, step 6 **restarts** it, so a changed wg0.conf is used. From 0.0.193:

- **DNS check first.** When wg0.conf's DNS setting goes through systemd-resolved (a `DNS =` line with systemd's `resolvconf`, or a resolvectl PostUp), it asks resolved for its status, with a 5-second limit. If there is no answer it restarts systemd-resolved once; if there is still none, the tunnel is not started, and the message says to reboot or remove the DNS setting. A `DNS =` line with no `resolvconf` at all is refused the same way.
- **Waits up to 90 seconds** for the start (it runs as a background job, so the page doesn't hang).
- **On failure** it shows the unit's state and the last 15 lines of its log since the button was pressed, instead of "command timed out".

From 0.0.200, after a successful start, step 6 also:

1. Waits up to 15 seconds for a handshake. If none comes it warns (check the Endpoint and that the tunnel is approved on the portal) and leaves the tunnel up.
2. Checks the tunnel server's route. If it goes into wg0, it adds the /32 route to the normal link straight away.
3. If `netcheck.service` is on the node, applies the netcheck fixes (7.5) that apply and aren't in place yet.
4. Sets the boot order (7.7, from 0.0.205).
5. Starts any enabled DVSwitch services that are stopped.

Each result is a line in the step's output. Convert's enable step does the same.

Verify the tunnel and the AllStar registration, under step 6, checks the tunnel itself from 0.0.212:

- the handshake's age, wg0's address, and that 44Net is routed through wg0;
- that the tunnel server doesn't loop into wg0, and that names resolve;
- each AllStar registration with the address it is registered as (as on the Overview card, 3.1);
- your public address through `connect.44net.cloud/myip`. On a split tunnel myip shows your home internet address, because connect.44net.cloud isn't a 44Net address; the handshake is what shows the tunnel works there.

> **CAREFUL** Starting the tunnel can change how the Pi answers the network. If you are connected over SSH or Cockpit from outside your home network, that session may drop. Do step 6 from home, or at a keyboard and screen.

7.5 The other buttons

- **Troubleshooting: bridge startup (netcheck)**, under step 6: when Analog_Bridge, MMDVM_Bridge or STFU won't start after the tunnel comes up. A read-only scan names each problem (netcheck's unit type, missing ordering after the network, bridges hard-tied to netcheck, the wait-online unit, its probe target), and buttons fix

each one: **fix netcheck's unit** (a drop-in: oneshot, after the network is online), **Loosen** a bridge, change the probe target and timeout, enable the wait-online unit, **Revert** all of it, and **Restart** netcheck and the bridges. Most likely needed on a full tunnel. From 0.0.200 step 6 and the DVSwitch installer apply these by themselves, except the probe target, which needs your input.
Loosen (from 0.0.201): a drop-in can't remove a dependency (systemd.unit(5): "dependencies can only be added in drop-ins"), so the earlier fix left netcheck in `Requires=`. 44helper now writes a copy of the bridge's unit to `/etc/systemd/system/<unit>`, first line `# 44helper netcheck override`, with netcheck moved to `Wants=` and the ordering kept. It counts as done only when `systemctl show` no longer lists netcheck in Requires. A unit already overridden in `/etc` by something else is refused. **Revert** deletes the copies, restores the original script (the first backup is kept), and disables the wait-online unit if 44helper enabled it.
From 0.0.211 the boot order doesn't rely on Loosen: it makes netcheck itself non-fatal with one drop-in (7.7), which also covers stfu and the packaged gateways that require netcheck in their own unit files. Loosen is still there for a single bridge; Revert also removes the non-fatal drop-in.

- **Firewall and port control**, at the end of the tab: the 44NetConnect zone and Quick port control (Chapter 4). On the Full Tunnel tab, the half tunnel card (Chapter 5) and Full tunnel with exceptions (Chapter 8) come before it.
- **Convert to Split / Full Tunnel**: switches an installed tunnel to the other kind, with a preview first.
- **Purge**: three steps: stop and disable the tunnel (and the fallback unit), remove `/etc/wireguard/wg0.conf`, and delete the 44NetConnect zone, which also drops wg0's attachment and every service and port in it. wg0.conf is copied to the instmon library first. From 0.0.201 step 2 is refused while the tunnel is up (wg-quick needs the file to stop it), and while the half tunnel or exceptions is permanent (use Remove permanent first).
- If one kind is installed and you open the other tab, its steps are locked until you press **Acknowledge and unlock**.

> **NOTE** A start that fails after exactly 25 seconds on `resolvconf -a wg0`, with "Failed to activate service 'org.freedesktop.resolve1': timed out", means systemd-resolved isn't answering on D-Bus. Run `sudo systemctl restart systemd-resolved`, check `timeout 5 resolvectl status` answers, and start again; or reboot; or take the `DNS =` line out.

7.6 Health and fixes

A card at the top of the Full and Split Tunnel tabs, from 0.0.199: the Overview's checks for the tunnel, each with a fix button. Like the firewall section, it exists once and moves to whichever tunnel tab you open. From 0.0.200 the install prevents these problems, so on a new tunnel the card should stay green; it is there for later changes, such as a file edited by hand or a service that fails at a reboot.

FINDING	FIX
Enabled DVSwitch services not running, or netcheck failed	**Start them**: starts each one and shows the last log lines of any that fail. **Apply netcheck fixes**: the netcheck card's fixes that apply and aren't in place yet.
Address wider than /32	**Change to /32**, when wg0.conf has exactly one 44Net address. Doesn't restart the tunnel. Check the portal first.
Old handshake, no keepalive	**Add keepalive (25 s)**: adds `PersistentKeepalive = 25` under `[Peer]` and applies it to the running tunnel, so no restart is needed.
Tunnel server routes into wg0, or wg0.conf has the older pin	**Fix the tunnel-server route**, or on a split tunnel **Switch to policy routing** (7.3).

Asterisk bound to the tunnel address (0.0.211)	**Set the boot order** (7.7), until the boot order lets Asterisk bind early. `bindaddr` in `/etc/asterisk/iax.conf` [general] (following `#include` and `#tryinclude`, as `ip` or `ip:port`) is wg0's address or another 44 address. On a split tunnel use `bindaddr=0.0.0.0`: IAX to nodes outside 44Net would leave through the router with a 44 source address. At boot Asterisk starts before wg0 has the address, so the bind fails until Asterisk restarts; the boot order adds a sysctl that lets it bind early.
Boot order not set (red when the tunnel isn't enabled at boot)	**Set the boot order** (7.7).
DNS revert in a PostDown line (0.0.201), or a PreDown line without `|| true` (0.0.211)	**Move to PreDown** (7.2), which also adds `|| true`. No restart needed.
DNS setting with systemd-resolved silent	**Restart systemd-resolved**, then checks it answers. **Set DNS to Off**: removes 44helper's DNS lines and any `DNS =` line.

- **Restart tunnel** is a separate button: the same DNS check and endpoint pin as step 6, a 90-second limit, and the log on failure.
- Every fix asks first and says what it changes. The node works out what to act on itself; the page can't name a service or a file.
- Fixes that write wg0.conf are refused while a half-tunnel or exceptions test is running.

> **CAREFUL** Restart tunnel can drop an SSH or Cockpit session from outside your home network. Use it from the LAN.

7.7 Boot order (step 6b)

From 0.0.205. After a reboot the tunnel has to come back by itself, and the DVSwitch services have to wait for it. Without this, they could start before the network, the tunnel or DNS were up: netcheck failed, the gateways couldn't reach their servers, and a tunnel start that failed at boot was never tried again. Step 6 sets it; the **6b. Boot order** card under step 6 shows what is in place, with **Set the boot order** and **Remove**; the Health card offers it when something is missing. It takes effect at the next boot; nothing running is restarted. The Boot tab (3.2) shows how the last boot went.

PIECE	WHAT IT DOES
`44helper-netready.service`	A oneshot that runs `/usr/local/sbin/44helper-netready` after `network-online.target` and the tunnel. It waits up to 180 seconds until there is a default route (one through wg doesn't count), the tunnel has a handshake (only when `wg-quick@wg0` is enabled and wg0.conf has an `Endpoint`) and names resolve, then finishes either way, so the services still start. From 0.0.211 it keeps real time: it works to a deadline and each lookup has a 3-second limit. Before, its timer counted sleeps; with DNS down each pass took about 13 seconds, systemd killed it at 210 seconds, and no boot-log line or late follow-up followed. Its journal line says which was missing: `journalctl -b -u 44helper-netready`.

Late follow-up `44helper-netready-late`	From 0.0.208. When netready gives up after 180 seconds, it starts this follow-up (with `systemd-run --collect`, never twice), which keeps watching for up to 20 more minutes. Once the route, the handshake and DNS are all there, from 0.0.211 it restarts netcheck (which restarts what requires it) and then only the enabled services that aren't running, Asterisk included, each after a `reset-failed` so a start limit can't block it.
Boot log `/var/lib/44helper/boot.log`	From 0.0.208. One line per boot from netready (ready, or what was missing and after how long), and one from the late follow-up. It keeps the last 20 lines, so the result survives a reboot even when System Optimization keeps the journal in memory only. The 6b card shows this boot's line, and the Health card warns when this boot wasn't ready in time.
DVSwitch drop-ins `44helper-boot.conf`	netcheck and each DVSwitch service that is installed (bridges, md380-emu, STFU, the gateways, ircDDBGateway, webproxy) start after it (`After=` / `Wants=`). From 0.0.208 this includes the capitalised names `Analog_Bridge`, `MMDVM_Bridge` and `STFU`. All but netcheck also restart 15 seconds after a failure; netcheck is retried 30 seconds after a failure, up to 10 times in 15 minutes.
netcheck non-fatal drop-in `netcheck.service.d/44helper-nonfatal.conf`	From 0.0.211. Repeats netcheck's `ExecStart` with a `-` prefix: netcheck still runs its check, but a failed check no longer fails the unit, so nothing that requires it is stopped. netready does the waiting. 44helper checks it with `systemctl show -p ExecStart` (`ignore_errors=yes`).
ASL3 update units	From 0.0.211 `asl3-update-astdb` and `asl3-update-nodelist` get the same ordering drop-in, so they wait for netready. On one node astdb had retried for 15 minutes and held the boot open.
Tunnel drop-in for `wg-quick@wg0`	`Restart=on-failure` every 20 seconds, with no limit, so a start that fails at boot (DNS or the network not ready, or the pin finding no route) is tried again. From 0.0.208 it first runs `/usr/local/sbin/44helper-timewait`: in the first 10 minutes after boot, when NTP is on, it waits up to 90 seconds for the clock to sync, then starts the tunnel either way. The Pi has no clock battery, and after a power cut its clock can start behind; the tunnel server drops handshakes with an older time than the last one it saw. From 0.0.212 the wait is **only for a full tunnel** (wg0.conf routes `0.0.0.0/0`): on a split tunnel the time sync goes over the LAN and the handshake works as soon as the clock is right, so the script exits at once (17 seconds saved on one node).
Asterisk early bind `/etc/sysctl.d/90-44helper-nonlocal-bind.conf`	From 0.0.211, only when `bindaddr` in iax.conf is wg0's address or another 44 address (7.6): `net.ipv4.ip_nonlocal_bind = 1`, so Asterisk can bind to that address at boot before wg0 has it.
Split tunnel routing	From 0.0.212 an older split tunnel is converted to policy routing (7.3); restart the tunnel afterwards.
NetworkManager hook `90-44helper-endpoint`	From 0.0.210, with NetworkManager only. Moves a pinned tunnel-server route when the network changes (7.3). From 0.0.212 needed only while wg0.conf still has a pin.
Enabled at boot	`wg-quick@wg0`, and the network's wait-online unit (`NetworkManager-wait-online` or `systemd-networkd-wait-online`), if they weren't.

Why "netcheck still required by ..." didn't clear before 0.0.211. The status checked all 13 DVSwitch units, but the fix only knew the three bridges: stfu's unit file is DVSwitch's own in `/etc/systemd/system`, so 44helper wouldn't copy over it, and the packaged gateways (ysfgateway and the others) require netcheck in their own unit files. Instead of copying unit files, Set the boot order (and Apply netcheck fixes) now writes the one non-fatal drop-in on netcheck itself. The card counts as done when no unit is held back by netcheck any more.

Set the boot order also applies the other netcheck fixes (7.5), and puts the tunnel server's route back on the normal link if it loops right now. Step 6's routing-loop check uses WireGuard's firewall mark (`ip route get … mark`), so a healthy full tunnel isn't reported as a loop.

Remove deletes the unit, the scripts, the drop-ins (the non-fatal one included), the sysctl file and the NetworkManager hook; the tunnel stays enabled at boot, and the policy-routing block stays in wg0.conf. The DVSwitch install step **Start DVSwitch after the network is ready** sets the DVSwitch part, and DVSwitch purge step 1 removes it (Chapter 10).

> **NOTE** wifimon's **tunnel watchdog** (5.35.0) is a safety net for what the boot order can't cover: five minutes after boot, and every minute after that, it restarts a tunnel that has no handshake and then the DVSwitch services. From 5.37.0 it fixes a routing loop at the first check (also during the wait after boot); from 5.38.0 it leaves the DVSwitch restart to the late follow-up while that is running, and allows a tunnel restart up to 240 seconds. Use wifimon 5.38.0 or later. See the user manual's wifimon chapter.

Full tunnel with exceptions TESTING

A card under the Full Tunnel tab's steps, from 0.0.194. Everything still goes through the tunnel, with two exceptions. It works like the half tunnel: temporary rules first, checks, then make it permanent.

8.1 The two exceptions

STEP	RULE	WHY
2. Keep replies from the LAN address on the LAN link	`from <LAN IP> lookup main · priority 1150`	A dashboard or SSH connection that comes in through the router's forward arrives on the LAN address. Without this, its replies go into the tunnel and the connection hangs.
3. Send the bypass list around the tunnel	`to <ip> lookup main · 1160` `ipproto udp dport <p> lookup main · 1161`	Optional. Host names, IPv4 addresses or UDP ports, space or comma separated. Host names are looked up when you press the button. The page suggests the DMR master from `MMDVM_Bridge.ini`. Press 3 again to change the list; an empty list is fine.

Both rules sit ahead of wg-quick's own full-tunnel rules (32764, 32765), and `lookup main` reaches the normal default route there. In test mode they are plain routing rules on the running tunnel: no restart.

> **NOTE** UDP router forwards, such as IAX to your home address, can't be kept. Asterisk picks its reply address from the route, which is the 44 address, so the reply doesn't match the original connection. Other nodes reach you at the 44 address instead. By default DV traffic also goes through the tunnel from the 44 address; BrandMeister and the reflectors log you in by ID, not by address.

8.2 The buttons

- **0. Check first** (read-only) blocks on: no wg0.conf, AllowedIPs not 0.0.0.0/0, a `Table =` line or `SaveConfig = true`, the tunnel not running from `wg-quick@wg0`, wg-quick's table 51820 rules not loaded, no normal default route or LAN address, the half tunnel in use. It lists what wg0's zone opens to the whole internet, and backs up wg0.conf as `wg0.conf.pre-fullx`.
- **1. Arm auto-undo**: the 15-minute timer (`44helper-fullxundo`).
- **2** and **3**: 8.1. **8. Undo now** and **9. Keep for now** as on the half-tunnel card.
- **10. Make permanent** (8.4) and **11. Remove permanent**.

8.3 The 12 checks

#	GROUP	CHECK	PASSES WHEN
1	Tunnel	Tunnel is up	Handshake under 3 minutes old.
2	Tunnel	Full-tunnel rules are loaded	wg-quick's table 51820 rules are in place.
3	Tunnel	Internet goes into the tunnel	`1.1.1.1` routes via wg0.
4	Tunnel	The internet sees your 44 address	`connect.44net.cloud/myip` reports the 44 address.
5	Exceptions	Replies from the LAN address stay on the LAN link	Traffic from the LAN address routes via the LAN link.

6	Exceptions	Bypass list uses the normal link	Every listed address and port routes via the LAN link (grey if the list is empty).
7	Normal use	DNS works	`allstarlink.org` resolves.
8	Normal use	LAN tools still respond	The tools answer and their replies to the LAN leave by the LAN link.
9–12	Services	DVSwitch, usrp2m17, registration, EchoLink	As on the half-tunnel card.

A red line on 5, 7 or 8 shows a banner recommending Undo now.

8.4 Make permanent and remove

- Unlocks after steps 2 and 3 and green or grey checks under 15 minutes old.
- Writes the rules as PostUp/PostDown lines between `# 44helper-fullx-begin` / `-end` in wg0.conf. They find the LAN address and look up the bypass hosts each time the tunnel starts. AllowedIPs isn't touched.
- Writes `remove_full_exceptions.sh` and a 0600 copy of wg0.conf to the chosen home folder and `/root`, plus a login reminder.
- Restarts the tunnel (after the DNS check) and runs the checks. If one fails, it takes everything out and restarts the test.
- **Remove permanent** and the script take the marked lines and the rules out. The tunnel stays a plain full tunnel; it isn't restarted.

> **CAREFUL** A bypass host is looked up when the tunnel starts. If DNS isn't working yet at boot, that host's rule is missing until the next tunnel restart.

Router Install, Nodes and Cloudflare

9.1 Router Install tab

For an OpenWrt-based router (GL.iNet models are recognised), driven over SSH with a key. Two models:

- **Model B, single address:** the router holds your one 44Net Connect address and forwards chosen ports to the Pi.
- **Model C, routed subnet:** a coordinator gives you a 44.x subnet; the router's LAN uses it, so the Pi and other devices get real 44Net addresses.

STEP	WHAT IT DOES
1. Router access	Method (none, ssh, api), host, user, SSH key path (for example `/etc/44helper/router_id_ed25519`) and poll interval. **Save**, then **Test connection / Re-probe**.
2. Firmware/model	OpenWrt? `uci`? GL.iNet model? Read-only.
2a. LuCI	Installs the OpenWrt web interface if it is missing (up to 15 minutes, with SSH keepalives).
3. Allocation type	Model C or Model B. The steps below change with it.
4. Capture config	Paste the WireGuard config (and the LAN subnet for Model C). The private key stays on the Pi.
5. Apply to router	Pushes the config with `uci`.
6. Router LAN IP *(C)*	Moves the router's LAN onto the subnet. **Your connection may drop**: renew the Pi's address or reboot it, then update the host in step 1.
7. Bring up the tunnel	`ifup wgclient`, then its status.
8. Pi's address	From the router's DHCP leases; Model C can add a static binding.
9. Forwards *(B)* / Firewall zone *(C)*	B: tick the services to forward to the Pi. C: the WireGuard zone forwards, with masquerade off.
10. Verify	B: outbound uses the 44.x address. C: a trace to portal.ampr.org. **Save known-good note** records the setup for the Actions Log.
Router: live port-forward table	Under the steps: the router's forwards, read over SSH, each with a **Remove** button. It moved here from the Firewall tab in 0.0.196.

> **NOTE** Model B's Verify can't prove other stations reach you from outside: 44helper can't test inbound from the internet. Ask another node to link to you.

9.2 Nodes tab

For more than one AllStar node on the same network. Each box needs its own IAX2 port, and the nodes need to know about each other.

CARD	SHOWS
This Box	The IAX2 `bindport` and whether it is in the recommended range (4560–4580). **Edit bindport** changes it.

Local Nodes	The nodes on this Pi, with their `rxchannel` and findings. **Edit** opens that node's line.
Peer Nodes	Other nodes on the network. **Wired** means `rpt.conf [nodes]` already has a line for it. Add one with node number, IP, port, label and, optionally, its 44helper address; a 🔗 **44helper** button then opens it (dimmed if it isn't answering).

Each finding has a **What to do** line. Saving an edit shows the line first, backs the file up, then shows the change.

9.3 Cloudflare tab

A Cloudflare Tunnel is an outgoing connection from the Pi to Cloudflare, which then serves a hostname you choose (such as `node.example.com`) down that connection. Nothing opens on your router. It carries web pages, not radio traffic.

METHOD	STEPS
token (persistent)	1. **Pre-flight** (read-only): is cloudflared installed, which pages run, how it fits with any WireGuard tunnel (none, split or full). 2. **Install cloudflared** from Cloudflare's repository. 3. **Install the connector service** (manual): run the `sudo cloudflared service install eyJ…` command from the Zero Trust dashboard over SSH, so the token never reaches 44helper's log. 4. **Verify** (read-only). 5. **Map a Public Hostname and add an Access policy** in the dashboard, for example `http://localhost:9999` for SysMon.
quick (test)	Pre-flight, install, then `cloudflared tunnel --url http://localhost:8989` over SSH: a random `*.trycloudflare.com` address, no account, gone at Ctrl-C.

From 0.1.3 step 1 (Pre-flight) also checks Cloudflare's apt source and signing key. Cloudflare changed its key on 30 October 2025 and the old one stopped working on 30 April 2026, so a node that installed cloudflared earlier fails every `apt update` until step 2 is run again, which replaces the key. Step 2 stops if another apt source file already names Cloudflare's repository. Both purges now remove the apt source and key.

Purge: uninstall the connector service, remove the package, the four clean-up steps, and optionally the stored credentials in `/etc/cloudflared`.

> **CAREFUL** A Cloudflare Tunnel **needs no inbound port**: cloudflared only connects out (to Cloudflare on port 7844), so the Ports tab and the firewall cards can't see what you published, and the mapping lives in Cloudflare where 44helper can't read it. Always put a **Cloudflare Access policy** (for example, email one-time codes for your address only) in front of every hostname. Never publish 44helper itself.

PART IV

Installers

Five tabs install, and remove, the radio software: ASL3, DVSwitch, SVXLink, M17 (USRP2M17) and Ampersand. Every step is listed here as it appears on the tab.

10. ASL3 and DVSwitch
11. SVXLink, M17 and Ampersand

ASL3 and DVSwitch

10.1 Before any install

- **Pick the right script.** Bookworm is Debian 12; Trixie is Debian 13. Check with `cat /etc/os-release`.
- **Pause the node on a Pi Zero 2 W:** quiet mode on the Update tab (Chapter 12), or instmon's Quiet System.
- **Watch the masked-service card** at the top of the tab (Chapter 2). Install buttons stay locked while it is red.
- **Afterwards**, check the Ports tab so nothing you installed is exposed.
- Cockpit (web admin, port 9090) has install steps on several tabs. Install it once; the others skip it. Its purge step is optional and never runs automatically.
- apt runs with `DEBIAN_FRONTEND=noninteractive` and `--force-confdef --force-confold`, so an upgrade keeps your edited config files instead of stopping to ask.

> **TIP** Several tabs end with **Save original configs as Restore point**. Run it once, after *all* the software you use is installed and set up (Chapter 13).

10.2 ASL3 tab

AllStarLink 3 from the official repo.allstarlink.org packages. Script: **Bookworm** or **Trixie** (same steps, different repository package: `asl-apt-repos.deb12_all.deb` or `deb13`).

#	INSTALL STEP	NOTES
1	Install prerequisites	wget, curl, git, firewalld, alsa-utils and the like.
2–4	Download ASL3 repo package · Install it · Refresh package index	The package lands in `/tmp`.
5	Install ASL3 core	`asl3`
6	Install Allmon3	
7–8	Install Cockpit · Enable Cockpit socket	
9	Install node database / TTS utilities	
10	*(Optional)* Full ASL3 Appliance branding/firewall defaults	`asl3-appliance`, `asl3-appliance-pi`; pulls in apache2.
11	Remove default Allmon3 account	
12	Set Allmon3 admin password	Fill in Username and Password. The password is on the command line briefly (process list, Actions Log).
13	Enable Allmon3 service	
14	Enable node database updater timer	`asl3-update-astdb.timer`
15	Configure node settings (asl-menu)	**Manual**: node number, callsign, radio interface. Full-screen: over SSH or in the built-in terminal.

16	Save original configs as Restore point	**Config**

#	PURGE STEP
1–5	Stop & disable ASL3-related services · Purge ASL3 and Allmon3 packages · Remove now-unused dependencies · Remove the ASL3 apt repository package · Clean up the downloaded file
6–9	The four clean-up steps (10.4)
opt.	Remove leftover node config (`/etc/asterisk`, your node settings) · Stop & purge Cockpit

10.3 DVSwitch tab

DVSwitch Server (Analog_Bridge, MMDVM_Bridge, md380-emu and the gateways) from dvswitch.org's repository.

BOOKWORM	TRIXIE
1. Download DVSwitch repo-install script 2. Run it (adds apt source + key) 3. Refresh package index 4. Install DVSwitch Server 5. Enable DVSwitch services at boot 6. Enable the digital-mode gateways at boot 7. **Start DVSwitch after the network is ready** 8–9. Install Cockpit · Enable Cockpit socket 10. Set Callsign / DMR ID / BrandMeister password 11. *(Optional)* Web-server clash: turn off the DVSwitch web dashboard 12. *(Optional)* … move it to another port 13. Save original configs as Restore point	1. **Check what's already here** (read-only) 2–4. Download and run the repo-install script, refresh 5. **Test install (changes nothing)**: a dry run; **if it fails, stop** 6. Install DVSwitch Server 7–8. Enable services and gateways at boot 9. **Start DVSwitch after the network is ready** 10–11. Cockpit 12. Callsign / DMR ID / BrandMeister password 13–14. *(Optional)* web-server clash fixes 15. Save Restore point

- **Start DVSwitch after the network is ready** (0.0.200 as "Stop netcheck from blocking boot"; boot order from 0.0.205): DVSwitch's netcheck runs at boot, and when it fails (often because the network isn't up yet) the bridges and gateways never start. This step applies the netcheck card's fixes (7.5) and the DVSwitch part of the boot order (7.7): netcheck, the bridges and the gateways wait for the network, the tunnel and DNS, and restart after a failure. Safe to run again; tunnel step 6 does the same.
- **Set Callsign / DMR ID / BrandMeister password** fills `MMDVM_Bridge.ini` from three fields and restarts the bridges. The DMR ID is your 7-digit ID plus a 2-digit suffix, for example `320123401`.
- **Web-server clash:** ASL3's apache2 (Allmon3) and DVSwitch's lighttpd dashboard both want port 80; step 1 reports a `CLASH`. Turn the DVSwitch dashboard off, or move it (8081 by default; ports used by the suite are refused). The move backs up `lighttpd.conf` once and checks the new config with `lighttpd -tt`. Both do nothing if there is no clash.

Purge (same for both scripts): from 0.0.200 it starts by removing 44helper's netcheck fixes (the card's Revert) and, from 0.0.205, the DVSwitch boot-order drop-ins, so nothing is left behind; then stop and disable the services, purge the packages, autoremove, remove the repository, the four clean-up steps; optional: leftover config, leftovers from the old third-party build script, Cockpit.

10.4 The four clean-up steps (every purge)

1. **Purge this tab's removed-but-not-purged packages.** Their leftover masks are what break the next install.

2. **Unmask this tab's own services.** Debian's intentional masks are never touched.
3. **Remove broken boot links** for this tab's services.
4. **Still left behind** (read-only): packages, masks, unit files and folders that remain, so nothing is left silently.

SVXLink, M17 and Ampersand

11.1 SVXLink tab

SCRIPT	STEPS
USRP fork (default)	1. Check for conflicts (read-only) · 2. Install build dependencies · 3. Clone the USRP fork (editable branch, default `usrp`) · 4. Create the svxlink user · 5. Configure & build (from source, no time limit) · 6. Check the USRP part was built · 7. Locate installed files · 8. Switch UsrpLogic on and set its ports (from the page: adds it to `LOGICS` and `LINKS`, sets the callsign and ports) · 9. Install & start the `svxlink-usrp` service (from the page) · 10. Set Callsign · 11–12. Cockpit
Bookworm / Trixie	Debian's `svxlink-server`: 1. Refresh · 2. Install SvxLink Server · 3. ALSA utilities · 4. Add the svxlink user to the audio and serial groups · 5. Set Callsign · 6. Enable at boot · 7–8. Cockpit. Sound card and PTT/COS stay at their defaults: set them for your hardware.

The USRP fork (DL1HRC) adds a UsrpLogic module so SvxLink can talk to the DVSwitch bridges over USRP. It installs under `/usr/local` (config in `/usr/local/etc`), so it can't collide with the package's files, but the two can't run at the same time.

> **CAREFUL** The fork is unofficial and not merged upstream. A reviewer found UsrpLogic crashes on a crafted short packet; keep it on loopback, never in a firewall zone or through a tunnel.

Purge covers both: stop and disable SvxLink (package and USRP build), purge the package, autoremove, the four clean-up steps; optional: leftover package config, the build's binaries and config, the hand-written unit, the cloned source, Cockpit.

11.2 M17 tab

USRP2M17-Selector (mjwgeek): the USRP2M17 bridge plus a small web page for picking M17 reflectors. It runs the project's own unmodified installer; its licence allows personal, non-commercial use only.

#	STEP	KIND
1–3	Install git · Install PHP for Apache · Clone USRP2M17-Selector	Run
4	Run the installer (asks for your callsign)	**Manual**, or the built-in terminal
5	`rpt.conf`: add the M17 node stanza (node 1917)	**Manual edit**
6	`modules.conf`: enable `chan_usrp`	**Manual edit**
7	`extensions.conf`: add the M17 extension	**Manual edit**
8	Restart Asterisk	Run
9	Save original configs as Restore point	Config
10	Install sudoers rule for the web page	Run
11–12	Cockpit	Run

The three Asterisk edits are shown as text, not run, because editing a live Asterisk config unattended is too risky. **Purge** stops and disables USRP2M17, removes the web page, the sudoers drop-in and the unit, then the four clean-up steps; optional: the clone directory, Cockpit. It **never** undoes the three edits.

> **TIP** Afterwards, SysMon's M17 tab checks the ports (32008 and 34008) and the `[1917]` node match, and the Config tab's Port check checks the USRP pair.

11.3 Ampersand tab

Ampersand is a separate, Asterisk-free AllStar/IAX node. Pick a **Product**:

PRODUCT	INSTALL	PURGE
Radio node (`amp-server`)	1. Check for conflicts · 2. Install required packages · 3. udev HID rule (COS/PTT) · 4. Download & unpack · 5. First run (foreground smoke test, built-in terminal) · 6. Install & start the service (its user and web UI port) · 7. (Optional) web UI password	Stop & disable · remove the unit · the binary · the staged files · clean-up; optional: node config and udev rule
Hub (`amp-hub`)	1. Check for conflicts · 2. Install net-tools, wget and libcurl4-gnutls-dev · 3. Download, unpack & run `install.sh` (a drop-in points the service at `/usr/etc/amp-hub.env`) · 4. Set the node number, password and IAX port in `/usr/etc/amp-hub.env` · 5. (Optional) other settings in that file · 6. Enable & start	Stop & disable · remove the unit · the service account · the staged files · clean-up; optional: the env file

> **CAREFUL** Ampersand **replaces** Asterisk; the two don't share a node. On a Pi already running ASL3 on IAX 4569/udp, give Ampersand its own node number and IAX port (the hub's form defaults to 4569), or use a separate Pi. The tab never stops Asterisk for you. The download date is pinned in the command box: change it if a newer build exists. Its web page (8080) should stay private.

PART V

Maintenance and configuration

Keeping the system healthy (Update, System Optimization) and keeping copies of the node's configuration (Config).

12. Update and System Optimization
13. The Config tab

Update and System Optimization

12.1 Quiet mode

Before a big upgrade or install, quiet mode pauses everything that competes for the SD card, in four tiers:

1. watchdogs and update timers (the Node Controller watchdog, monit, the ASL3 node-list and database timers, man-db);
2. background disk users (the performance recorder, PackageKit, the manual-page indexer);
3. the suite's web tools (SysMon, the Dashboard, instmon, the SvxLink and M17 dashboards, and lighttpd, the DVSwitch web page);
4. the radio services (Asterisk, the bridges, md380-emu, SvxLink, ircDDBGateway, STFU).

Turn off restarts exactly what was paused, in reverse. 44helper, wifimon, SSH, Cockpit, networking and the firewall are never touched. The card shows the commands first. If 44helper restarts while quiet mode is on, it is restored in the background. From 0.0.212 the quiet-mode state records the boot it was turned on in: quiet mode left on before a **reboot** is cleared when 44helper starts, because the reboot already started everything, instead of restarting services that are still waiting for the network.

> **CAREFUL** The node is **off the air** while quiet mode is on, and it stays on until you turn it off: no 30-minute auto-restore, unlike instmon's Quiet System. It can't be turned on while a package job holds apt's locks. If something doesn't come back, the card shows the likely cause and offers **Reboot node**.

12.2 Update tab

- **Package Lookup — Installed** lists installed packages; **Version Policy** shows installed and candidate versions of one package.
- **Upgradable Packages**, **Held Packages**, and a banner when a **reboot is required**. **Re-check** refreshes them.

#	STEP	ALTERNATIVES
1	Refresh package index (`apt update`)	
2	Fix missing/broken packages (`dpkg --configure -a`, `apt install -f`). Safe any time.	Is apt still running? (read-only) · Reconfigure half-installed packages
3	Upgrade packages (`apt upgrade`)	Full upgrade: can *remove* packages. Read the Upgradable list first.
4	Autoremove unused packages	Autoremove + purge configs (not reversible)
5	Clean local package cache	Autoclean (keeps recent downloads)
6–7	*Optional:* Webmin (port 10000, HTTPS) · Ajenti 2	Remove each
8	*Optional:* fail2ban (SSH jail)	Check status · Remove
9	**Package recovery 1: check** (read-only)	
10	**Package recovery 2: finish half-installed packages**	

11 **Package recovery 3: re-install broken packages**

Steps that change packages have **Run in quiet mode**; leave it ticked on a Pi Zero 2 W.

> CAREFUL **fail2ban:** edit the `ignoreip` line before step 8. It ships with `192.168.1.0/24`, which may not be your network; add your tunnel subnet if you manage the node over 44Net. A ban lasts 10 minutes and covers SSH only. To lift one: `fail2ban-client set sshd unbanip <ip>`.

After a cut-off upgrade

1. Step 2's **Is apt still running?**: wait until nothing is.
2. **Step 9**: lists problems. If any, **don't reboot yet**: a half-installed kernel or firmware package can stop the Pi from starting.
3. **Step 10**: finishes ordinary packages first, kernel and firmware last, at low priority; stops at the first failure.
4. **Step 11**, only if 9 or 10 says so: reinstalls from the copy on the Pi, or downloads just that package.
5. **Step 9** again: nothing listed. Now reboot.

12.3 System Optimization tab

Cuts SD-card wear. Each card shows the current state, an action, a **Revert** that puts back exactly what was there before this tab changed it, the commands, and **Copy**.

CARD	WHAT IT DOES	UNDO
Root fstab — noatime	Stops a write on every file read. The candidate fstab is checked with `findmnt --verify`, then `mount -o remount,noatime /`.	Revert; one backup of `/etc/fstab` kept
journald — volatile	Keeps the system log in memory (drop-in `/etc/systemd/journald.conf.d/44helper-sysopt.conf`).	Revert
rsyslog	Stops and disables the second, text-file logger.	Revert
mandb — Disable rebuild	Stops the periodic manual-page index rebuild (`man-db.timer`).	Revert
mandb — Remove package	Removes `man-db`, after showing what apt would remove.	Reinstall, **needs internet**
Swap — restore	Only on a node where an older 44helper turned swap off or removed `dphys-swapfile`. **Revert** turns swap back on; **Reinstall** puts the package back (needs internet).	—

> CAREFUL 0.0.195 removed the swap Disable and Uninstall cards: turning swap off ran low-RAM nodes out of memory. Keep swap on. And with journald volatile and rsyslog off, **no log survives a crash or power cut**; remember that before looking for why a node rebooted.

The Config tab

The safety net for the node's configuration: copies of every config file, restored in one go, wiring carried from one Pi to another, and checks that callsigns, IDs and ports agree everywhere.

13.1 The ideas

TERM	MEANS
System	A group of files restored together: **ASL3**, **DVSwitch**, **USRP2M17** or **Phone**. Only installed systems are saved.
Restore point	The original config, saved **once** after your installs are finished.
Save point	A named copy of the current files, any time.
Automatic copy	Taken before every restore, preset, callsign change or undo.
Preset	Only a node's *wiring* (private nodes, ports, extensions), no identity. Today: **Travel Node**.

Up to **10** points in all. When a new automatic copy would go over, the oldest automatic copies go first; named save points are never removed for you.

13.2 The cards

CARD	WHAT IT DOES
Restore point	**Save Restore point** once; saving again only adds systems installed since. Each changed file has **View diff**. **Retake Restore point** replaces it, keeping the old one as a save point.
Restore	Pick a point in **From**, tick the systems, **Restore selected**. The current files are saved first; the output lists each file as written, unchanged or skipped. **Delete** removes a save point or automatic copy.
Restart services	After a change: buttons that restart exactly the services those files belong to. Nothing restarts until you press one.
Finish: reinstall and reboot	Reinstalls the Node Controller tools from instmon's library (instmon first, each one put back if its new copy doesn't start) and reboots once instmon, the Dashboard and Asterisk are healthy. Don't power off while it runs.
Create save point	A name of up to 40 characters (letters, numbers, spaces, `_ . -`).
Travel Node preset	The private nodes (1000–1999) in `rpt.conf` and their `[nodes]` lines, their extensions, `chan_usrp`, the USRP2M17 and Analog_Bridge `[USRP]` ports. Never callsigns, IDs or passwords. **Preview**, then **Save as Travel Node**; apply it from Restore on any node. Bridge nodes get the upstream USRP2M17 values, and it warns before breaking EchoLink or a working USRP pair.

Add private node (from 0.1.4)	Adds a private node (1000–1999) to `rpt.conf` the way the Dashboard's Phone tab and Node Talker do: a stanza, its `[nodes]` line on this node's IAX port and, if ticked, an extension in `[radio-secure]`. A blank number picks the next free one; 1921 is refused because Node Talker adds it. The type is Plain or USRP bridge. **Preview** shows every change; **Add node** saves an automatic copy first and doesn't restart Asterisk. Nodes it added are listed with **Preview remove** and **Remove**; other nodes are never touched. The Travel Node preset leaves Node Talker's 1921 alone. With Allmon3 installed, adding or removing a node also updates `/etc/allmon3/allmon3.ini` (from 0.1.5).
Identity	Checks callsigns, IDs and passwords agree across every file (masked; **Reveal** needs the root password). Chips: **matches**, **differs** (red), **some copies empty**, **not set**, **no public node**, **one copy**.
Set the callsign everywhere TESTING	On the Identity card's Callsign line: 3–7 letters and digits with at least one digit. Preview first, an automatic save point, then every callsign field (rpt.conf ID text, EchoLink, MMDVM_Bridge, `var.txt`, ircDDBGateway, the YSF/P25/NXDN gateways, USRP2M17, the Dashboard's and 44helper's config). Nothing restarts: restart the Dashboard first. The ircDDB and DPlus accounts must exist under the new call.
Digital-mode gateways	**Enable and start at boot** for ircDDBGateway, YSF/FCS, P25, NXDN and STFU, so every mode is ready after boot. **Undo** puts them back.
Port check	UDP ports two programs both listen on; whether each gateway mirrors MMDVM_Bridge; **USRP pairs** for every `rxchannel = USRP/127.0.0.1:A:B`. **Verify with SysMon** compares with `sysmon.py --usrp-pairs` (SysMon 6.13.67.16-pi02w / 6.13.82 or later). **Check what is listening now** adds the live picture: reachable ports, gateway command ports, broad firewall ranges.

13.3 Common tasks

Undo a change that broke the node

1. Restore: pick the save point or automatic copy from before the change.
2. Tick the system involved and press Restore selected.
3. Restart what Restart services lists, or run Finish: reinstall and reboot.

Copy a travel setup to a new Pi

1. On the working node: Travel Node preset, Preview, Save as Travel Node.
2. On the new node, with the same instmon library: install the software (Part IV), then Restore → Travel Node.
3. Identity: set this node's callsign everywhere, and fill in its IDs and passwords until every group matches.

Copies are kept in `/etc/asl_dvs/instmon_library/config_restore/` (automatic copies under `auto/`), so instmon's Backup & Export ZIP includes them. Each restored file is written through a temporary file (mode 600, no symlink following), then given the saved owner and mode and swapped into place.

PART VI

Reference

14. Troubleshooting
15. Files, services, timers and routes

Troubleshooting

14.1 A step says NOT RUN

1. Read the reason in its output: masked services, or broken packages.
2. Masks: use the card at the top of the tab. Broken packages: follow **What to do next** to the Update tab's recovery steps.
3. Come back with the link at the top of the Update tab and run the step again.

14.2 The tunnel won't start

```
systemctl status wg-quick@wg0 --no-pager
journalctl -u wg-quick@wg0 -b -l --no-pager | tail -30
sudo wg show wg0
```

THE LOG STOPS AT	CAUSE AND FIX
`resolvconf -a wg0`, 25 s, "org.freedesktop.resolve1: timed out"	systemd-resolved isn't answering on D-Bus. `sudo systemctl restart systemd-resolved`; if `timeout 5 resolvectl status` still hangs, reboot. Or remove the `DNS =` line.
`wg setconf`, "Name or service not known... Trying again"	The Endpoint's host name can't be looked up: check DNS with `getent ahostsv4 <host>`.
a `PostUp` line	That hook failed; the error is on the next line.
`nft -f` / `iptables-restore`	wg-quick's full-tunnel firewall step failed: `nftables` or `iptables` is missing.

Up but no handshake, or the dashboard says **idle**: first run `ip -4 route get <endpoint>`. If it says `dev wg0`, the tunnel server is routed into its own tunnel: Split tab step 4b, **Switch to policy routing** (section 7.3). Otherwise check the key and the Endpoint on connect.44net.cloud, and that UDP out to the Endpoint isn't blocked. Your `Address` should be what the portal shows (a single 44Net Connect address is a /32).

14.3 I lost access after a tunnel change

- A half-tunnel or exceptions **test** takes itself out after 15 minutes, or at a reboot.
- A **permanent** one: log in at a keyboard and screen, or over SSH from the LAN, and run `~/remove_half_tunnel.sh` or `~/remove_full_exceptions.sh`.
- A plain full tunnel and router-forwarded SSH: use the LAN, or the full tunnel with exceptions (Chapter 8).

14.4 DVSwitch or M17 stopped after the tunnel came up

1. Split tab: a pasted full config sends DNS into the tunnel. Step 4b, Repair AllowedIPs.
2. Bridges won't start at boot: the Health and fixes card's **Apply netcheck fixes**, or the netcheck card under step 6 (Chapter 7). From 0.0.200 step 6 and the DVSwitch installer do this by themselves.
3. Full tunnel: outbound block rules (Full tab 4b) or a master that refuses the 44 address. Try the half tunnel, or put the master on the exceptions bypass list.

14.5 The tunnel or DVSwitch doesn't come back after a reboot

1. Open the **Boot** tab (3.2) and read the findings. To ask for help, press **Copy boot report** (secrets are removed) or run `sudo python3 /usr/local/bin/44helper.py --boot-report`. The saved copies keep the last 5 boots even when the journal is in memory only.
2. Health card: if it offers **Set the boot order**, press it, then reboot (7.7).
3. Split tunnel with a routing loop, or card 4b showing **Older routing**: **Switch to policy routing** (7.3). The tunnel server then stays on the normal route whichever adapter is up, so WiFi changing adapter can't loop it.
4. WiFi adapter set to **Keep off** in wifimon: use wifimon 5.38.0 or later. Earlier versions let NetworkManager bring the adapter up at boot and then took it down, which took the tunnel's route with it. From 5.38.0 NetworkManager never uses a Keep-off adapter.
5. For the detail: `cat /var/lib/44helper/boot.log` (last 20 boots), or `journalctl -b -u wg-quick@wg0 -u 44helper-netready -u 44helper-netready-late -u netcheck --no-pager | head -80`. The netready line says whether the network, the tunnel handshake or DNS was late.
6. A full tunnel gets no handshake for the first minutes after a power cut: the clock was behind. The tunnel waits for the clock (7.7); check `timedatectl` shows `NTP service: active`.
7. "netcheck still required by ..." on the 6b card with 0.0.210 or older: update 44helper and Set the boot order again (7.7).
8. wifimon's tunnel watchdog restarts a tunnel with no handshake after five minutes and then the DVSwitch services; its card on the wifimon page shows what it did.

14.6 AllStar is registered at the wrong address

The Overview's **AllStar registration** card (3.1) and Verify (7.4) show the address the AllStarLink server sees. On a split tunnel that is your public internet address; on a full or half tunnel it should be the 44 address.

- **A private address:** the server sees you behind NAT it can't reach; check the router.
- **Asterisk bound to the 44 address** (Health card, 7.6): with `bindaddr` set to wg0's address in `/etc/asterisk/iax.conf`, IAX fails at boot until Asterisk restarts, and on a split tunnel IAX to nodes outside 44Net leaves through your router with a 44 source address. On a split tunnel set `bindaddr=0.0.0.0` and restart Asterisk. If you keep the 44 address, Set the boot order so the bind works at boot.

14.7 The Pi runs out of memory

Check swap is on: `swapon --show`. If an older 44helper turned it off, use **Swap — restore** (Chapter 12) or `sudo systemctl enable --now dphys-swapfile`.

Files, services, timers and routes

Files

PATH	HOLDS
`/usr/local/bin/44helper.py`	the program (a link into `/usr/local/lib/asl_dvs/44helper/`); `/opt/44helper/asl_dvs_m17_44helper.py` is a second link for older tools
`/etc/44helper/44helper.conf`	settings: server, identity, tunnels, router access, nodes
`/etc/44helper/44helper_guard_keep.json`	services you chose to keep masked
`/etc/44helper/44helper_quiet_state.json`	what quiet mode paused, and from 0.0.212 the boot it was turned on in
`/etc/44helper/terminal.enable`	terminal switch
`/run/44helper/sessions.json`	logins (cleared at reboot)
`/etc/wireguard/wg0.conf`	the tunnel config (mode 600); marker blocks `44helper-dns`, `-mtu`, `-routing` (split policy routing, 0.0.212), `-half`, `-fullx`, and `-endpoint-pin` (older split tunnels only)
`/etc/wireguard/wg0.conf.pre-replace`	the config before the last **Replace the config** (0.0.212)
`/etc/firewalld/zones/44NetConnect.xml`	the 44Net zone
`/etc/asl_dvs/instmon_library/config/`	wg0.conf backups: `wg0.conf`, `wg0.conf.pre-halftunnel`, `wg0.conf.pre-fullx`
`/etc/asl_dvs/instmon_library/config_restore/`	the Config tab's Restore point, save points and automatic copies
`/run/44helper-half/ · /run/44helper-fullx/`	test state, undo scripts and logs (cleared at reboot)
`/etc/44helper/halftunnel.json · fullx.json`	what Make permanent saved
`/etc/sysctl.d/90-44helper-halftunnel.conf`	loose return-path check on wg0
`/etc/sysctl.d/90-44helper-nonlocal-bind.conf`	`net.ipv4.ip_nonlocal_bind = 1`, only when Asterisk binds to a 44 address (boot order, 0.0.211)
`/usr/local/sbin/44helper-netready · 44helper-timewait`	the boot order's network wait and clock wait (7.7)
`/etc/systemd/system/<unit>.d/44helper-boot.conf`	boot-order drop-ins: the tunnel, netcheck, the DVSwitch services and (0.0.211) the ASL3 update units
`/etc/systemd/system/netcheck.service.d/44helper-nonfatal.conf`	netcheck's check made non-fatal (0.0.211); replaces the copied bridge unit files for the boot order
`/etc/NetworkManager/dispatcher.d/90-44helper-endpoint`	moves a pinned tunnel-server route when the network changes (0.0.210); legacy, needed only while wg0.conf has a pin
`/run/44helper-endpoint-wg0`	the pinned tunnel server's address (older split tunnels only)

`/var/lib/44helper/boot.log`	one line per boot from netready and the late follow-up, last 20 lines
`/var/lib/44helper/boots/`	saved boot reports (0.0.209): one per boot, newest 5, 64 KB each, mode 0600
`/etc/profile.d/44helper-halftunnel.sh` · `44helper-fullx.sh`	login reminders
`~/remove_half_tunnel.sh` · `~/remove_full_exceptions.sh`	removal scripts (also in `/root`), with 0600 wg0.conf copies

Services and timers

UNIT	PURPOSE
`44helper.service`	the page, port 9997
`wg-quick@wg0` · `44net-tunnel.service`	the tunnel, or the ARDC fallback unit
`44helper-halfundo.timer` · `44helper-fullxundo.timer`	the 15-minute auto-undo (transient, made by `systemd-run`)
`44helper-job-*`	package and install jobs, run as their own systemd units
`44helper-netready.service`	the boot order's wait for the network, tunnel and DNS (7.7)
`44helper-netready-late`	the late follow-up (transient, made by `systemd-run --collect`)

Routing

PRIORITY / TABLE	OWNER
1100–1105, table 4444	half tunnel
10045 (`not fwmark 0x4445`), table 4445	split tunnel policy routing (0.0.212 used 4444/10044; 0.0.213 and later 4445/10045); WireGuard marks its own packets with `FwMark = 0x4445`
1150, 1160, 1161	full tunnel with exceptions
32764, 32765, table 51820	wg-quick's own full-tunnel rules

Web routes

Every route needs the login cookie; every POST also needs an `X-Requested-With` header. Each tab has `/api/<tab>/script` (its step list) and `/api/<tab>/action`; jobs are polled at `/api/action_status?job_key=…`. The tunnel cards use `/api/firewall/half_status`, `/api/fullx/status` and `/api/firewall/action`. The Boot tab uses `/api/boot/status`, `/api/boot/report?format=json|text`, `/api/boot/section?id=…` and `/api/boot/snapshot?name=…` (3.2).